

Assignment No. 2: Test Plan and Test Case Writing

Project: Online Book Store

1. Test Plan:

1.1 Test Plan Identifier

Plan ID: TP-OBS-2025-001

Version: 1.0

Date: 01/05/2026

Author: QA Team

1.2 Test Plan Overview

This test plan defines the testing strategy, approach, resources, and schedule for validating the Online Book Store application. The plan encompasses all functional requirements (FR-1 through FR-13), non-functional requirements (NFR-1 through NFR-13), and user stories (US-1 through US-20) documented in Assignment No. 1.

1.3 Testing Objectives

1. Validate that all 13 functional requirements operate correctly
2. Verify that all 13 non-functional requirements meet specified criteria
3. Ensure all 20 user stories deliver expected business value
4. Confirm acceptance criteria scenarios execute successfully
5. Identify defects before production deployment
6. Validate system performance under specified load conditions
7. Ensure security requirements (encryption, PCI DSS compliance, session timeout) function properly

1.4 Scope of Testing

1.4.1 In-Scope Features (by Sprint)

Sprint 1: Account Management

1. User registration functionality (US-1)
2. User login functionality (US-2)
3. Session timeout after 1 hour inactivity (US-18)
4. Password encryption storage (NFR-1)

Sprint 2: Search and Discovery

1. Search by title, author name, and category (FR-1, US-3)
2. Filter by price range, language, publication year, rating (FR-3, US-4)
3. Sort by price ascending and descending (FR-4, US-5)

4. Search result loading within 3 seconds (NFR-4, US-19)

Sprint 3: Book Information

1. Complete book details display (FR-2, US-6)
2. User reviews and ratings display (FR-6, US-7)
3. Book detail page loading within 3 seconds (NFR-5)

Sprint 4: Wishlist and Shopping Cart

1. Add and remove books from wishlist (FR-5, US-8)
2. Add and remove items from shopping cart (FR-8, US-9)
3. Grand total calculation (FR-8, US-10)

Sprint 5: Checkout and Payment

1. Secure payment processing via PCI DSS compliant gateway (FR-7, US-11)
2. Save address and payment information (FR-10, US-12)
3. Order confirmation email delivery (FR-12, US-13)
4. Checkout completion within 10 seconds (NFR-6)
5. Email delivery within 5 minutes (NFR-11)

Sprint 6: Order Management

1. Order tracking with status updates (FR-9, US-14)
2. Order history retention (FR-11, US-15)
3. Real-time status display (NFR-12)

Sprint 7: Device Support and Performance

1. Responsive design across devices (FR-13, NFR-8, US-16)
2. Legible font sizes across devices (NFR-9)
3. Tappable buttons across devices (NFR-10)
4. 1000+ concurrent user support (NFR-13, US-20)
5. 99.9% system availability (NFR-7)

1.4.2 Out-of-Scope Items

1. Social media integration (Facebook, Google login)
2. Mobile native applications (iOS/Android apps)
3. Affiliate marketing tracking
4. International shipping calculations
5. Digital book downloads
6. Subscription services
7. Gift card functionality
8. Bulk order discounts

1.5 Test Strategy

1.5.1 Testing Levels

Unit Testing (Developer Responsibility)

1. Individual component testing
2. Password hashing validation
3. Price calculation logic

Integration Testing (QA Responsibility)

1. Database connectivity
2. Payment gateway integration
3. Email service integration
4. Session management

System Testing (QA Responsibility)

1. End-to-end user journeys
2. Cross-browser compatibility
3. Cross-device responsiveness

Acceptance Testing (QA + Product Owner)

1. User story validation
2. Acceptance criteria verification

Regression Testing (QA Responsibility)

1. Existing functionality verification after each sprint

Performance Testing (QA Responsibility)

1. Load testing with 1000 concurrent users
2. Response time validation
3. Availability monitoring

Security Testing (QA Responsibility)

1. Password encryption verification
2. Session timeout validation
3. Payment information security

1.5.2 Testing Types by Category

Test Type	Coverage	Sprint
Smoke Testing	Critical paths (login, search, add to cart, checkout)	All Sprints
Regression Testing	Previously working features	Sprints 2-7
Negative Testing	Invalid inputs, edge cases	Sprint 1-6
Security Testing	Password encryption, session timeout, PCI DSS	Sprints 1,5,7
Performance Testing	Response times, concurrent user	Sprints 2,3,5,7
Usability Testing	Responsive design, font legibility, button tappability	Sprint 7
Integration Testing	Payment gateway, email service	Sprint 5

1.6 Test Environment Specifications

1.6.1 Hardware Environment

Device Type	Specifications	Screen Resolution
Desktop/Laptop	Windows 11, 16GB RAM, Intel i7	1920x1080, 1366x768
Desktop/Laptop	macOS Ventura, 8GB RAM, M1 chip	2560x1600
Tablet	iPad Pro 11-inch	1194x834
Tablet	Samsung Galaxy Tab S7	1600x2560
Mobile	iPhone 14 Pro	2556x1179
Mobile	Samsung Galaxy S23	2340x1080
Mobile	Google Pixel 7	2400x1080

1.6.2 Software Environment

Browser	Versions	Devices
Google Chrome	Latest 3 versions	Desktop, Mobile
Mozilla Firefox	Latest 3 versions	Desktop
Safari	Latest 2 versions	Desktop, Mobile, Tablet
Microsoft Edge	Latest 2 versions	Desktop

1.6.3 Network Environment

1. Wired connection: 100 Mbps
2. Wi-Fi: 5 GHz and 2.4 GHz
3. 4G/5G cellular: Mobile network testing
4. Throttled connection: 3G simulation (for performance edge cases)

1.6.4 Test Servers

Server	Purpose	URL
Development	Active feature testing	dev.bookstore.internal
Staging	Integration and regression testing	staging.bookstore.com
Production clone	Performance testing	perf.bookstore.com

1.7 Test Data Requirements

1.7.3 Book Catalog Data

Field	Example Values
Titles	Harry Potter and the Sorcerer's Stone, The Great Gatsby, Python Programming, Clean Code
Authors	J.K. Rowling, F. Scott Fitzgerald, Robert C. Martin

Categories	Fiction, Non-Fiction, Technology, Fantasy, Mystery
Price Range	\$9.99 to \$49.99
Languages	English, Spanish, French, German
Publication Years	2020, 2021, 2022, 2023, 2024
Ratings	1 to 5 stars
Availability	In Stock, Out of Stock, Pre-order

1.7.4 Test User Accounts

User Type	Email	Password	Purpose
New User	newuser@test.com	Test@1234	Registration testing
Existing User	existing@test.com	Pass#5678	Login testing
Frequent Buyer	frequent@test.com	Buy#9012	Saved address/payment testing
Long Session	session@test.com	Session@3456	Timeout testing

1.7.5 Test Payment Information

Card Type	Card Number	Expiry	CVV	Purpose
Visa	4111 1111 1111 1111	12/28	123	Successful payment
Mastercard	5555 5555 5555 4444	10/27	456	Successful payment
Declined Card	4000 0000 0000 0002	12/28	123	Payment failure testing
Invalid Card	1234 5678 9012 3456	01/25	789	Validation testing

1.8 Entry and Exit Criteria

1.8.1 Entry Criteria for Each Testing Sprint

1. Development completed for assigned user stories
2. Code peer reviewed and merged
3. Unit tests passed (80% minimum coverage)
4. Build deployed to staging environment
5. Test data prepared and seeded
6. Test cases reviewed and approved

1.8.2 Exit Criteria for Each Sprint

1. All planned test cases executed
2. Critical defects (Severity 1) closed
3. High severity defects (Severity 2) closed or accepted by product owner
4. Test execution report generated
5. Regression testing passed for previous functionality

1.8.3 Final Release Exit Criteria

1. 100% of test cases executed (minimum 95% pass rate)
2. Zero Severity 1 or Severity 2 open defects
3. Performance criteria met (3 second load times, 10 second checkout)
4. Security validation passed (encryption, PCI DSS compliance)
5. Responsiveness verified across 6+ device/browser combinations
6. Load testing passed for 1000 concurrent users
7. Availability monitoring confirms 99.9% uptime

1.9 Suspension and Resumption Criteria

1.9.1 Suspension Criteria

Testing will be suspended when:

1. Critical defect prevents further testing (e.g., login unavailable)
2. Environment becomes unstable (frequent crashes)
3. Payment gateway sandbox is unavailable
4. Database corruption occurs
5. Build deployment fails

1.9.2 Resumption Criteria

Testing will resume when:

1. Critical defect is fixed and verified
2. Environment is stabilized by system administrator

3. Payment gateway service is restored
4. Database is restored from backup
5. Successful build is redeployed

1.10 Deliverables

Deliverable	Format	Due	Responsibility
Test Plan Document	PDF/Word	Sprint 1 start	QA Lead
Test Cases (20+)	Testworthy	Daily during Sprint	QA Engineer
Test Execution Reports	Testworthy Export	After each Test Run	QA Engineer
Defect Reports	Jira/Testworthy	As discovered	QA Engineer
Regression Test Suite	Testworthy	Sprint 2-7	QA Engineer
Performance Test Report	PDF	End of Sprint 7	Performance Engineer
Security Validation Report	PDF	End of Sprint 7	Security Tester
Test Summary Report	PDF	End of each Sprint	QA Lead

1.11 Risks and Mitigation

Risk	Probability	Impact	Mitigation Strategy
Payment gateway sandbox instability	Medium	High	Maintain backup test card set; coordinate with payment provider
Environment downtime	Low	High	Schedule testing during stable hours; maintain local test environment
1000 concurrent user load not achievable	Medium	Medium	Use cloud-based load testing tools; simulate gradually

Mobile device fragmentation	High	Medium	Prioritize top 10 device/browser combinations; use responsive design testing tools
Session timeout conflicts with long test cases	Medium	Low	Reset session before each test case; document timeout in test steps
Email delivery delays in test environment	Low	Medium	Use email testing service (Mailtrap); implement retry mechanism

1.12 Test Execution Schedule

Sprint 1 (Week 1): Account Management

Activity	Day	Duration
Test case creation for US-1, US-2, US-18	Day 1	4 hours
Smoke testing	Day 2	2 hours
Functional testing	Day 2-3	8 hours
Negative testing (invalid passwords, wrong credentials)	Day 3	4 hours
Security testing (password encryption, session timeout)	Day 4	4 hours
Defect verification	Day 4	2 hours
Regression testing	Day 5	2 hours
Report generation	Day 5	2 hours

Sprint 2 (Week 2): Search and Discovery

Activity	Day	Duration
Test case creation for US-3, US-4, US-5, US-19	Day 1	6 hours
Smoke testing (existing + new features)	Day 2	3 hours
Functional testing (search, filters, sort)	Day 2-3	10 hours

Performance testing (3 second load time)	Day 3	3 hours
Negative testing (no results, invalid filters)	Day 4	3 hours
Regression testing (Sprint 1 features)	Day 4	3 hours
Defect verification	Day 5	2 hours
Report generation	Day 5	2 hours

Sprint 3 (Week 3): Book Information

Activity	Day	Duration
Test case creation for US-6, US-7	Day 1	4 hours
Smoke testing	Day 2	2 hours
Functional testing (book details, reviews)	Day 2-3	8 hours
Performance testing (3 second page load)	Day 3	2 hours
Regression testing (Sprint 1-2 features)	Day 4	4 hours
Defect verification	Day 4	3 hours
Report generation	Day 5	2 hours

Sprint 4 (Week 4): Wishlist and Shopping Cart

Activity	Day	Duration
Test case creation for US-8, US-9, US-10	Day 1	5 hours
Smoke testing	Day 2	2 hours
Functional testing (wishlist, cart operations)	Day 2-3	10 hours
Negative testing (empty cart, duplicate adds)	Day 3	3 hours
Regression testing (Sprint 1-3 features)	Day 4	5 hours
Defect verification	Day 5	2 hours

Report generation	Day 5	2 hours
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Sprint 5 (Week 5): Checkout and Payment

Activity	Day	Duration
Test case creation for US-11, US-12, US-13	Day 1	6 hours
Smoke testing	Day 2	3 hours
Functional testing (checkout flow)	Day 2-3	8 hours
Integration testing (payment gateway, email)	Day 3	4 hours
Security testing (PCI DSS compliance)	Day 4	4 hours
Performance testing (10 second checkout)	Day 4	2 hours
Regression testing (Sprint 1-4 features)	Day 5	4 hours
Report generation	Day 5	2 hours

Sprint 6 (Week 6): Order Management

Activity	Day	Duration
Test case creation for US-14, US-15	Day 1	4 hours
Smoke testing	Day 2	2 hours
Functional testing (order tracking, history)	Day 2-3	8 hours
Real-time status validation	Day 3	3 hours
Regression testing (Sprint 1-5 features)	Day 4	5 hours
Defect verification	Day 5	2 hours
Report generation	Day 5	2 hours

Sprint 7 (Week 7): Device Support and Performance

Activity	Day	Duration
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Test case creation for US-16, US-17, US-20	Day 1	4 hours
Cross-device testing (6+ devices)	Day 2-3	12 hours
Performance load testing (1000 concurrent users)	Day 3	4 hours
Security testing (encryption final validation)	Day 4	3 hours
Full regression suite (all 20 US)	Day 4-5	10 hours
Defect verification	Day 5	3 hours
Final test summary report	Day 5	3 hours

1.13 Roles and Responsibilities

Role	Name	Responsibilities
QA Lead	[Name]	Test plan creation, team coordination, final sign-off
QA Engineer 1	[Name]	Account, search, book details testing
QA Engineer 2	[Name]	Wishlist, cart, checkout testing
QA Engineer 3	[Name]	Order management, device, performance testing
Automation Engineer	[Name]	Performance test scripting, regression automation
Security Tester	[Name]	Password encryption, PCI DSS validation
Product Owner	[Name]	Acceptance criteria approval, defect prioritization
Development Lead	[Name]	Defect fixes, environment support

1.14 Tools

Tool	Purpose
Testworthy	Test case management, execution, reporting
Jira	Defect tracking, sprint management
Selenium	Automated regression testing (selected test cases)

JMeter	Load testing for 1000 concurrent users
BrowserStack	Cross-browser and cross-device testing
Mailtrap	Email testing for order confirmations
Stripe Sandbox	Payment gateway testing
Chrome DevTools	Performance measurement, responsive testing

2. Detailed Test Cases

2.1 Test Case Naming Convention

Format: TC-[Module][Sequential Number]

Example: TC-ACC-001 (Account module, case 001)

Test Suite 1: Account Management (ACC)

Classification: Smoke, Regression

TC-ACC-001: Create New User Account

Field	Detail
User Story	US-1: As a new user, create account and login successfully
Acceptance Criteria	Account gets created, user auto-logged in, redirected to dashboard
Priority	High
Test Type	Positive, Functional
Classification Tags	Smoke, Regression, Automation

Preconditions:

- 1. User is not registered in the system
- 2. Sign-up page is accessible (URL: bookstore.com/signup)
- 3. Browser cookies are enabled

Test Data:

Field	Value
Email Address	newcustomer2025@testmail.com

Password	SecurePass#123
Confirm Password	SecurePass#123

Test Steps:

Step	Action	Expected Result
1	Navigate to sign-up page	Sign-up form displays with email, password, confirm password fields and "Create Account" button
2	Enter valid email address in email field	Email address appears in field without error messages
3	Enter valid password in password field	Password appears as dots/bullets for security
4	Enter same password in confirm password field	Confirmation matches password field
5	Click "Create Account" button	System processes registration request
6	Wait for response	Account creation confirmation displays within 3 seconds
7	Check login status	User is automatically logged in without additional login step
8	Check current page URL	Browser URL changes to bookstore.com/dashboard
9	Verify dashboard elements	Dashboard displays user email, account menu, and bookstore navigation

Postconditions:

User account exists in database with encrypted password
 User session is active
 User can access protected pages without re-authentication

Negative Variations:

TC-ACC-002: Existing email address → Error "Email already registered"
 TC-ACC-003: Password mismatch → Error "Passwords do not match"
 TC-ACC-004: Invalid email format → Error "Enter valid email address"
 TC-ACC-005: Weak password → Error "Password must contain 8+ chars, number, special char"

TC-ACC-006: Login with Existing Credentials

Field	Detail
User Story	US-2: Existing user logs in with saved credentials
Acceptance Criteria	User logged in successfully and redirected to dashboard
Priority	High
Test Type	Positive, Functional
Classification Tags	Smoke, Regression, Automation

Preconditions:

User account exists with email: existing@test.com, password: Pass#5678

User is not currently logged in

Login page is accessible (URL: bookstore.com/login)

Test Data:

Field	Value
Email Address	existing@test.com
Password	Pass#5678

Test Steps:

Step	Action	Expected Result
1	Navigate to login page	Login form displays with email, password fields and "Login" button
2	Enter registered email address	Email appears in email field
3	Enter correct password for the account	Password appears as dots/bullets
4	Click "Login" button	System validates credentials
5	Wait for authentication	Loading indicator appears briefly

6	Check login success	Page redirects to dashboard within 3 seconds
7	Verify dashboard display	Dashboard shows user name from account, recent orders (if any), and navigation menu
8	Check session persistence	Refresh page - user remains logged in
9	Verify account menu	Account menu shows options: My Profile, My Orders, Wishlist, Logout

Postconditions:

User session created with session ID stored
Session timeout timer starts at 60 minutes
User can access all authenticated features

Negative Variations:

TC-ACC-007: Wrong password → Error "Invalid email or password"
TC-ACC-008: Non-existent email → Error "Invalid email or password"
TC-ACC-009: Empty fields → Error "Email and password required"
TC-ACC-010: Case sensitivity in email → Login succeeds (email case insensitive)

TC-ACC-011: Session Timeout After Inactivity

Field	Detail
User Story	US-18: Session timeout after 1 hour inactivity
Acceptance Criteria	User logged out after 1 hour, sees session expired message
Priority	High
Test Type	Security, Timing
Classification Tags	Security, Regression

Preconditions:

User is logged in with active session
System session timeout configured for exactly 3600 seconds (1 hour)
Current time is known for timing verification

Test Data:

Field	Value
User Email	session@test.com
Current Time	Record start timestamp

Test Steps:

Step	Action	Expected Result
1	Log in with valid credentials	User successfully logs in, dashboard displayed
2	Record current timestamp	Start time noted: [HH:MM:SS]
3	Navigate to any page (e.g., book catalog)	Page loads normally, session active
4	Perform no actions for 55 minutes	User remains idle, no interaction with website
5	After 55 minutes, attempt to access a page	User can still access pages (session still active)
6	Wait additional 6 minutes (61 minutes total)	Additional time passes with no activity
7	Attempt to navigate to another page	System detects expired session
8	Observe page response	Page redirects to login page
9	Check for session expired message	Message displays: "Your session has expired. Please log in again"
10	Try to access dashboard URL directly	Redirected to login page again, cannot bypass
11	Log in again	Login succeeds, new session created

Postconditions:

Old session invalidated and cannot be reused

New session created upon re-login

Session timeout reset to 60 minutes

Negative Variations:

TC-ACC-012: Activity at 59 minutes → Session remains active

TC-ACC-013: Activity after timeout but before page refresh → Next request triggers logout

TC-ACC-014: Background tabs without focus → Inactivity still counts (no mouse/keyboard events)

Test Suite 2: Search and Discovery (SRCH)

Classification: Smoke, Regression

TC-SRCH-001: Search Books by Title

Field	Detail
User Story	US-3: Search books by title, author name, or category
Acceptance Criteria	Search results visible within 3 seconds, showing title, author, price, cover
Priority	High
Test Type	Positive, Functional
Classification Tags	Smoke, Regression, Automation, Performance

Preconditions:

Book catalog contains "Harry Potter and the Sorcerer's Stone" by J.K. Rowling

Book catalog contains "Harry Potter and the Chamber of Secrets" by J.K. Rowling

Book catalog contains "Harry Potter and the Prisoner of Azkaban" by J.K. Rowling

At least 50 other books in catalog for result diversity

User is on homepage

Test Data:

Field	Value
Search Query	Harry Potter
Search Type	Title

Test Steps:

Step	Action	Expected Result
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1	Locate search bar on homepage	Search bar visible at top of page with placeholder text "Search by title, author, or category"
2	Type "Harry Potter" into search bar	Text appears as typed
3	Click search button (magnifying glass icon) or press Enter	Search request submitted
4	Start timer immediately	Timer begins at 0 seconds
5	Wait for results to display	Results page loads
6	Stop timer when results appear	Time recorded: [value]
7	Verify load time is within 3 seconds	Load time ≤ 3 seconds
8	Check each result card	Each result displays: book title, author name, price, cover image
9	Verify all Harry Potter books appear	All 3 Harry Potter books present in results
10	Verify order of results	Results sorted by relevance (default)
11	Click on a result	Navigates to book detail page
12	Perform search with partial title "Potter"	Returns Harry Potter books plus any other books with "Potter" in title

Postconditions:

Search results cached for 5 minutes to improve repeated search performance

Search term saved in recent searches (if user logged in)

Negative Variations:

TC-SRCH-002: No results → Message "No books found matching '[query]'"

TC-SRCH-003: Special characters "Harry@# \$" → Returns results ignoring special characters or shows error

TC-SRCH-004: Extremely long query (500 characters) → Truncated or returns no results gracefully

TC-SRCH-003: Filter Books by Price Range and Rating

Field	Detail
User Story	US-4: Filter books by price range, language, publication year, rating
Acceptance Criteria	Only books matching criteria are visible
Priority	High
Test Type	Positive, Functional
Classification Tags	Regression

Preconditions:

User performed search for "fiction" returning 50+ results

Results include books with various prices (\$5 to \$50)

Results include books with various ratings (1 to 5 stars)

Filter panel is visible on search results page

Test Data:

Filter	Selection
Price Range	\$10.00 - \$20.00
Rating	4 stars and above

Test Steps:

Step	Action	Expected Result
1	Execute search for "fiction"	Search results page displays with 50+ books
2	Locate filter panel on left side or top	Filter panel shows options: Price Range, Language, Publication Year, Rating
3	Find price range filter	Price range shows min/max inputs or slider
4	Enter minimum price \$10.00	\$10.00 appears in min field
5	Enter maximum price \$20.00	\$20.00 appears in max field

6	Click "Apply" or wait for auto-apply	Filter is applied
7	Find rating filter	Rating shows stars (1-5) or checkboxes
8	Select "4 stars and above"	Rating filter applied
9	Observe result count	Result count decreases from 50+ to smaller number
10	Examine each displayed book	Every book has price between \$10.00 and \$20.00
11	Examine each displayed book rating	Every book has rating of 4.0 or higher
12	Remove price filter	Results expand again
13	Remove rating filter	Results expand to original count

Postconditions:

Filter selections persist while user stays on search page

URL updates with filter parameters for sharing (e.g., ?price=10-20&rating=4)

Negative Variations:

TC-SRCH-004: Min price greater than max price → Shows error or swaps values

TC-SRCH-005: No books match filters → Message "No books match selected filters. Try broader criteria."

TC-SRCH-006: Apply language filter "Spanish" → Only Spanish language books display

TC-SRCH-007: Sort Books by Price (Low to High)

Field	Detail
User Story	US-5: Sort books in ascending and descending order based on prices
Acceptance Criteria	Cheapest book shown first, most expensive book shown last
Priority	High
Test Type	Positive, Functional
Classification Tags	Regression

Preconditions:

Search results contain books with varying prices

Minimum price in results: \$5.99

Maximum price in results: \$49.99

Sort dropdown is available on search results page

Test Data:

Field	Value
Search Query	programming
Sort Option	Price: Low to High

Test Steps:

Step	Action	Expected Result
1	Execute search for "programming"	Search results display 30+ books
2	Locate sort dropdown	Sort dropdown visible near results, default showing "Relevance"
3	Click sort dropdown	Dropdown expands showing options: Relevance, Price: Low to High, Price: High to Low, Rating, Newest
4	Select "Price: Low to High"	Sort selection changes
5	Wait for results to reorder	Results refresh with new sort order
6	Identify price of first result	First book price appears (should be lowest in set)
7	Verify first book price	First book price equals \$5.99 or lowest available
8	Identify price of last result on page	Last visible book price appears
9	Verify ascending order	Each subsequent book has price greater than or equal to previous book

10	Select "Price: High to Low"	Results reverse order
11	Verify first book is most expensive	First book price equals \$49.99 or highest available
12	Verify descending order	Each subsequent book has price less than or equal to previous book
13	Re-select "Relevance"	Returns to original default ordering

Postconditions:

Sort preference remembered during session

Sort parameter reflected in URL

Negative Variations:

TC-SRCH-008: Apply sort with filters → Both sort and filters work together correctly

TC-SRCH-009: Books with same price → Order by secondary criteria (title alphabetically)

Test Suite 3: Book Information (BOOK)

Classification: Smoke, Regression

TC-BOOK-001: View Detailed Book Information

Field	Detail
User Story	US-6: See complete book details including description, price, author, cover, rating, availability
Acceptance Criteria	Detail page loads within 3 seconds showing all required information
Priority	High
Test Type	Positive, Functional
Classification Tags	Smoke, Regression, Performance

Preconditions:

Book exists in catalog with complete metadata

Test book: "Clean Code" by Robert C. Martin

Book has all fields populated

Test Data:

Field	Expected Value
Book Title	Clean Code: A Handbook of Agile Software Craftsmanship
Author	Robert C. Martin
Price	\$39.99
Description	Text of at least 200 characters
Cover Image	Image URL pointing to valid image
Rating	4.7 out of 5 stars
Availability	In Stock

Test Steps:

Step	Action	Expected Result
1	Navigate to book detail page for "Clean Code"	Page URL: bookstore.com/books/clean-code
2	Start timer when navigation begins	Timer starts at 0 seconds
3	Wait for page to fully load	Page loading indicator completes
4	Stop timer when content visible	Load time recorded
5	Verify load time ≤ 3 seconds	Load time passes or fails requirement
6	Locate book title	Title displayed prominently at top of page
7	Verify title matches	"Clean Code: A Handbook of Agile Software Craftsmanship" displayed
8	Locate author information	Author name displayed near title
9	Verify author	"Robert C. Martin" displayed
10	Locate price	Price displayed with currency symbol

11	Verify price	"\$39.99" or equivalent displayed
12	Locate cover image	Book cover image displays
13	Verify image loads	Image loads without broken icon, alt text present
14	Locate description section	Description text block visible
15	Verify description content	Description contains at least one paragraph of text about book
16	Locate rating display	Star rating or numeric rating visible
17	Verify rating	"4.7 out of 5 stars" or similar displayed
18	Locate availability status	Availability indicator visible (green "In Stock" or red "Out of Stock")
19	Verify availability	"In Stock" displayed
20	Scroll entire page	No layout breaks, all sections visible

Postconditions:

Page may be cached for subsequent views

Page view counter increments (if implemented)

Negative Variations:

TC-BOOK-002: Book with missing cover image → Default placeholder image displayed

TC-BOOK-003: Book out of stock → "Out of Stock" message, "Add to Cart" button disabled or replaced with "Notify Me"

TC-BOOK-004: Invalid book ID → 404 page with "Book not found" message and link to browse

TC-BOOK-005: View User Reviews and Ratings

Field	Detail
User Story	US-7: Read reviews from other buyers
Acceptance Criteria	All reviews visible with text and rating
Priority	Medium
Test Type	Positive, Functional

Classification Tags	Regression
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Preconditions:

Book has at least 5 existing reviews from other users

Reviews include various ratings (3, 4, 5 stars)

Reviews include text comments

User is on book detail page

Test Data:

Review Attribute	Example Values
Reviewer Name	JohnD, BookLover99
Rating	5 stars, 4 stars, 3 stars
Review Text	"Excellent book, highly recommend", "Good but overpriced"
Date	2025-03-15, 2025-02-28

Test Steps:

Step	Action	Expected Result
1	Navigate to book detail page for book with reviews	Book detail page loads
2	Scroll down past book description	Reviews section is below description
3	Locate reviews section heading	Heading "Customer Reviews" or "Reviews" displayed
4	Verify overall rating summary	Summary shows average rating (e.g., "4.2 out of 5 stars") and total count (e.g., "128 ratings")
5	Locate individual reviews	Each review appears in separate container
6	Verify first review elements	Review shows: reviewer name, star rating, review text, date
7	Check reviewer name	Name displayed (or "Anonymous" if hidden)
8	Check star rating	Correct number of stars filled for that review

9	Check review text	Text content displayed as written
10	Check date	Date in readable format (e.g., "March 15, 2025")
11	Verify all 5 reviews visible	Scroll through - all 5 reviews appear
12	Check pagination if more than 10 reviews	"Load More" button or page numbers appear
13	Click "Load More" if present	Additional reviews load without page refresh
14	Verify sort options (if present)	Options: "Most Recent", "Highest Rating", "Lowest Rating", "Most Helpful"
15	Change sort to "Highest Rating"	Highest rated reviews appear first

Postconditions:

New reviews can be added (if user purchased book)

Reviews cached for performance

Negative Variations:

TC-BOOK-006: Book with no reviews → Message "No reviews yet. Be the first to review this book!" with option to write review

TC-BOOK-007: Review with no text but rating only → "No written review" placeholder displayed

TC-BOOK-008: Offensive review content (if moderation implemented) → Flagged and hidden or pending approval

Test Suite 4: Wishlist (WISH)

Classification: Regression

TC-WISH-001: Add Item to Wishlist

Field	Detail
User Story	US-8: Save books to wishlist for future consideration
Acceptance Criteria	Item saved to wishlist, button text changes
Priority	Medium
Test Type	Positive, Functional

Classification Tags	Regression
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Preconditions:

User is logged into account

User is on book detail page for "The Great Gatsby"

Book is not already in user's wishlist

Wishlist icon/count is visible in navigation bar

Test Data:

Field	Value
Book Title	The Great Gatsby
Author	F. Scott Fitzgerald

Test Steps:

Step	Action	Expected Result
1	Log into user account	User logged in, dashboard displayed
2	Navigate to book detail page for "The Great Gatsby"	Book detail page loads
3	Locate "Add to Wishlist" button	Button visible near "Add to Cart" button, typically heart icon or text button
4	Note initial button text/icon	Button shows "Add to Wishlist" or outline heart icon
5	Click "Add to Wishlist" button	Button click registers
6	Observe button change	Button text changes to "Remove from Wishlist" or heart icon becomes filled
7	Check wishlist count in navigation bar	Wishlist count increments by 1
8	Verify confirmation message	Toast or banner message: "The Great Gatsby added to your wishlist" appears briefly

9	Navigate to another page	Navigate to different book
10	Return to "The Great Gatsby" detail page	Button still shows "Remove from Wishlist"
11	Log out and log in again	Button still shows "Remove from Wishlist" (persistence confirmed)

Postconditions:

Database record created linking user ID to book ID

Wishlist persisted after logout/login

Negative Variations:

TC-WISH-002: Add book already in wishlist → Button shows "Remove from Wishlist", clicking removes it

TC-WISH-003: Add to wishlist when not logged in → Redirect to login page or prompt "Please log in to save to wishlist"

TC-WISH-004: Add book that is out of stock → Still adds to wishlist successfully

TC-WISH-005: View Wishlist Items

Field	Detail
User Story	US-8: Find saved items easily when ready to buy
Acceptance Criteria	Wishlist page shows all saved books, clicking book navigates to detail page
Priority	Medium
Test Type	Positive, Functional
Classification Tags	Regression

Preconditions:

User is logged in

User has 3 books in wishlist: "The Great Gatsby", "1984", "To Kill a Mockingbird"

Wishlist count shows 3

Test Data:

Wishlist Item	Expected Display
The Great Gatsby	Title, author, price, cover thumbnail, remove button
1984	Title, author, price, cover thumbnail, remove button
To Kill a Mockingbird	Title, author, price, cover thumbnail, remove button

Test Steps:

Step	Action	Expected Result
1	Log into user account	Dashboard displayed
2	Locate "My Wishlist" button/link	Option in account menu or navigation bar
3	Click "My Wishlist"	Navigation to wishlist page
4	Verify page URL	URL contains "/wishlist" or "/account/wishlist"
5	Check page title	Page heading "My Wishlist" displayed
6	Verify all 3 books appear	All 3 books visible on page
7	For each book, check displayed information	Each book shows: cover image, title, author, price
8	Verify cover images load	All cover thumbnails display properly
9	Verify book titles are clickable	Mouse cursor changes to pointer on hover
10	Click on first book title	Navigation to book detail page for that book
11	Return to wishlist page	Use browser back or wishlist link
12	Click on remove button (X or trash icon) for one book	Remove action triggered
13	Verify book removed	Book no longer appears in wishlist
14	Verify wishlist count updates	Navigation bar wishlist count decreases to 2
15	Verify confirmation	Message "[Book title] removed from wishlist" appears

16	Check empty wishlist message after removing all books	Message "Your wishlist is empty. Start adding books you love!" with link to browse
----	---	--

Postconditions:

Removed books deleted from wishlist database table

Removed books can be added again

Negative Variations:

TC-WISH-006: Empty wishlist → Shows empty state message and "Browse Books" button

TC-WISH-007: Book removed from catalog but in wishlist → Shows "Book no longer available" message

TC-WISH-008: Price changed on wishlist → Shows current price (not price at time of adding)

Test Suite 5: Shopping Cart (CART)

Classification: Smoke, Regression

TC-CART-001: Add Item to Shopping Cart

Field	Detail
User Story	US-9: Add items to cart for purchasing multiple books
Acceptance Criteria	Item added to cart, cart count updates
Priority	High
Test Type	Positive, Functional
Classification Tags	Smoke, Regression, Automation

Preconditions:

User is logged in

Cart is currently empty (cart count shows 0)

User is on book detail page for "Python Crash Course" (price: \$29.99)

Book is in stock

Test Data:

Field	Value
-------	-------

Book Title	Python Crash Course
Quantity Available	10+

Test Steps:

Step	Action	Expected Result
1	Log into user account	User logged in
2	Navigate to book detail page for "Python Crash Course"	Book detail page loads
3	Locate "Add to Cart" button	Button visible, typically green or primary color
4	Note initial cart icon count	Cart icon shows "0" or no badge
5	Click "Add to Cart" button	Button click registers
6	Observe cart count update	Cart icon badge increments to "1"
7	Verify confirmation message	Toast or banner: "Python Crash Course added to cart" appears
8	Click "Add to Cart" again for same book	Cart count increments to "2" (if quantity not restricted)
9	Navigate to different book "Learning Python"	Different book detail page loads
10	Click "Add to Cart" for second book	Cart count increments to "3"
11	Click cart icon	Cart page opens showing both items

Postconditions:

Cart items stored in database or session

Cart persists after logout/login if user is logged in

Cart saved for 30 days for guest users (if implemented)

Negative Variations:

TC-CART-002: Add to cart when not logged in → Items added to guest cart, cart persists via cookies, prompts to login at checkout

TC-CART-003: Add out of stock book → "Add to Cart" button disabled, shows "Out of Stock"

TC-CART-004: Add book with maximum quantity reached → Error message "Maximum quantity (10) reached for this item"

TC-CART-005: View Cart and Grand Total

Field	Detail
User Story	US-10: See grand total of items before final purchase
Acceptance Criteria	Cart page shows all items and grand total
Priority	High
Test Type	Positive, Functional
Classification Tags	Regression

Preconditions:

User is logged in

Cart contains 3 items:

Item 1: "Python Crash Course" - \$29.99, quantity 1

Item 2: "Learning Python" - \$39.99, quantity 2

Item 3: "Fluent Python" - \$49.99, quantity 1

Test Data:

Item	Price	Quantity	Subtotal
Python Crash Course	\$29.99	1	\$29.99
Learning Python	\$39.99	2	\$79.98
Fluent Python	\$49.99	1	\$49.99
Grand Total			\$159.96

Test Steps:

Step	Action	Expected Result
1	Log into user account with pre-populated cart	User logged in

2	Click cart icon in navigation bar	Cart page loads
3	Verify page URL	URL contains "/cart"
4	Check page heading	Heading "Shopping Cart" displayed
5	Verify all 3 items displayed	All 3 books visible in cart list
6	For each item, verify displayed information	Shows: cover thumbnail, title, price, quantity selector, remove button
7	Verify item 1 details	Title "Python Crash Course", price "\$29.99", quantity "1"
8	Verify item 2 details	Title "Learning Python", price "\$39.99", quantity "2"
9	Verify item 3 details	Title "Fluent Python", price "\$49.99", quantity "1"
10	Locate subtotal for each item	Each item shows subtotal (price $\times$ quantity)
11	Verify item 1 subtotal	\$29.99 displayed
12	Verify item 2 subtotal	\$79.98 displayed
13	Verify item 3 subtotal	\$49.99 displayed
14	Locate grand total section	Grand total displayed prominently, typically at bottom right
15	Verify grand total calculation	\$159.96 displayed
16	Change quantity of item 2 from 2 to 3	Use quantity dropdown or +/- buttons
17	Verify item 2 subtotal updates	\$119.97 (39.99 $\times$ 3)
18	Verify grand total updates	\$199.95 (29.99 + 119.97 + 49.99)

Postconditions:

Cart total recalculated after any quantity change

Cart persists with updated quantities

Negative Variations:

TC-CART-006: Empty cart $\rightarrow$ "Your cart is empty" message with "Continue Shopping" button

TC-CART-007: Cart with discount code applied → Grand total shows original total, discount, and final total

TC-CART-008: Item price changes after adding to cart → Cart shows updated price, notification "Price changed for [item]"

TC-CART-009: Remove Item from Cart

Field	Detail
User Story	US-9, US-10: Remove items and see updated total
Acceptance Criteria	Item removed, grand total updates
Priority	High
Test Type	Positive, Functional
Classification Tags	Regression

Preconditions:

User is logged in

Cart contains 2 items:

Item A: "The Pragmatic Programmer" - \$49.99

Item B: "Clean Architecture" - \$44.99

User is on cart page

Test Data:

Item	Price	Subtotal
The Pragmatic Programmer	\$49.99	\$49.99
Clean Architecture	\$44.99	\$44.99
Grand Total		\$94.98

Test Steps:

Step	Action	Expected Result
1	Navigate to cart page	Cart page displays both items, grand total \$94.98

2	Locate remove button for "Clean Architecture"	Remove button (trash icon or "Remove" text) visible next to item
3	Click remove button	Confirmation dialog may appear (optional)
4	Confirm removal if prompted	Item removal processed
5	Observe cart content	"Clean Architecture" no longer appears in cart
6	Verify remaining item	Only "The Pragmatic Programmer" remains
7	Verify grand total updates	Grand total changes to \$49.99
8	Verify cart count in navigation	Cart badge shows "1" instead of "2"
9	Verify confirmation message	Message "Clean Architecture removed from cart" appears
10	Remove remaining item	Click remove for "The Pragmatic Programmer"
11	Verify empty cart state	Message "Your cart is empty" and "Continue Shopping" button appear

Postconditions:

Removed items deleted from cart database table

User can add removed items again

Negative Variations:

TC-CART-010: Remove item then undo (if implemented) → Item returns to cart with original quantity

TC-CART-011: Remove item during checkout → Cart updates, checkout page reflects changes

Test Suite 6: Checkout and Payment (CHK)

Classification: Smoke, Regression

TC-CHK-001: Successful Order Placement with Saved Information

Field	Detail
User Story	US-11, US-12: Pay securely using card and save address/payment information

Acceptance Criteria	Order placed successfully, checkout completes within 10 seconds
Priority	Critical
Test Type	Positive, Functional, Integration
Classification Tags	Smoke, Regression, Security

Preconditions:

User is logged in as frequent buyer "frequent@test.com"

User has saved address: "123 Main St, Anytown, AN 12345"

User has saved payment method: Visa ending 1111

Cart contains 1 item: "Design Patterns" - \$54.99

Payment gateway sandbox is available

Test Data:

Field	Value
Shipping Address	Saved address: 123 Main St, Anytown, AN 12345
Payment Method	Saved Visa ending 1111
Order Total	\$54.99

Test Steps:

Step	Action	Expected Result
1	Add item to cart and navigate to cart page	Cart shows "Design Patterns" with price \$54.99
2	Click "Proceed to Checkout" button	Checkout page loads
3	Start timer	Timer begins at 0 seconds
4	Verify checkout page elements	Checkout shows sections: Shipping Address, Payment Method, Order Summary

5	Locate saved addresses	Address selector shows saved address "123 Main St..."
6	Select saved address	Address selected, radio button or checkbox checked
7	Locate payment methods	Payment selector shows saved Visa ending 1111
8	Select saved payment method	Payment method selected
9	Review order summary	Summary shows item, price, tax (if any), total \$54.99
10	Click "Place Order" button	Order submission processes
11	Wait for order confirmation	Processing indicator displays
12	Stop timer when confirmation received	Time recorded
13	Verify checkout completes within 10 seconds	Time $\leq$ 10 seconds
14	Verify order confirmation page	Confirmation page displays: Order Number, Order Details, "Thank You" message
15	Capture order number	Order number recorded (e.g., ORD-12345)
16	Verify order number format	Order number follows expected pattern

Postconditions:

Order record created in database

Cart becomes empty

Inventory quantity decreases by 1 for purchased book

Negative Variations:

TC-CHK-002: No saved address → User must enter new address manually

TC-CHK-003: No saved payment → User must enter card details manually

TC-CHK-004: Secure Payment Processing via PCI DSS Gateway

Field	Detail
User Story	US-11: Pay using credit, debit, or Visa card with secure method

Acceptance Criteria	Payment processed through PCI DSS compliant gateway, card info not stored in plain text
Priority	Critical
Test Type	Security, Integration
Classification Tags	Security

Preconditions:

User is logged in

User is on checkout page

User has items in cart

Payment gateway sandbox configured

Test Data:

Field	Value
Card Number	4111 1111 1111 1111 (Visa)
Expiry Date	12/28
CVV	123
Cardholder Name	TEST USER

Test Steps:

Step	Action	Expected Result
1	Navigate to checkout page	Payment section displays card entry fields
2	Enter new payment method (not saved)	Card form appears
3	Enter Visa card number 4111111111111111	Card number accepts input, may show card type icon (Visa)
4	Enter expiry date 12/28	Date accepted
5	Enter CVV 123	CVV field masks input with dots

6	Enter cardholder name "TEST USER"	Name accepted
7	Inspect page source using browser dev tools	Card number not present in HTML source as plain text (should be in iframe or encrypted)
8	Submit order and complete payment	Order processes
9	Check network traffic in dev tools	Card details not sent in plain text in API requests
10	Verify payment processed by gateway	Order confirmation received
11	Check database (if accessible)	Card numbers stored as tokens, not plain text
12	Verify PCI DSS compliance indicators	Payment page shows security badges (e.g., "PCI Compliant", lock icon)
13	Check for HTTPS	URL shows https://, padlock icon in address bar

Postconditions:

Payment token stored instead of raw card data

Transaction recorded in payment gateway

Negative Variations:

TC-CHK-005: Declined card (4000000000000002) → Error message "Payment declined. Please use another card."

TC-CHK-006: Invalid CVV → Error "Invalid CVV code"

TC-CHK-007: Expired card → Error "Card has expired"

TC-CHK-008: Receive Order Confirmation Email

Field	Detail
User Story	US-13: Receive email confirmation after placing order
Acceptance Criteria	Email received within 5 minutes containing order number, items, total
Priority	High
Test Type	Integration, Timing

Classification Tags	Regression
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Preconditions:

User account has email: confirm@test.com (using Mailtrap or test inbox)

User has successfully placed order TC-CHK-001

Email service is operational

Current time recorded at order placement

Test Data:

Field	Expected Content
Recipient	confirm@test.com
Subject	"Order Confirmation #[Order Number]"
Order Number	ORD-12345 (from placement)
Items	Design Patterns - \$54.99
Total	\$54.99

Test Steps:

Step	Action	Expected Result
1	Record order placement timestamp	Time A = [HH:MM:SS]
2	Complete order placement	Order confirmation page displayed
3	Navigate to email test inbox (Mailtrap)	Inbox loaded
4	Refresh inbox every 30 seconds	Check for new email
5	Wait for email to arrive	Email appears in inbox
6	Record email arrival timestamp	Time B = [HH:MM:SS]
7	Calculate delivery time	Delivery time = Time B - Time A

8	Verify delivery ≤ 5 minutes	Delivery time passes or fails requirement
9	Verify email sender	From address: orders@bookstore.com or noreply@bookstore.com
10	Verify subject line	Subject contains "Order Confirmation" and order number
11	Open email	Email content displays
12	Verify order number in email	Order number matches ORD-12345
13	Verify items listed	"Design Patterns - \$54.99" appears
14	Verify total amount	"\$54.99" appears
15	Verify shipping address	"123 Main St, Anytown, AN 12345" appears
16	Verify estimated delivery date	Date (order date + shipping days) appears
17	Verify tracking link	Link to order tracking page appears (active after shipment)
18	Verify customer service contact	Support email or phone number visible

Postconditions:

Email stored in user's order history

Resend email option available in account settings

Negative Variations:

TC-CHK-009: Email sent to incorrect address → Order confirmation still appears on site, user can update email in profile

TC-CHK-010: Email marked as spam → User checks spam folder and finds email, can mark as "Not Spam"

Test Suite 7: Order Management (ORD)

Classification: Regression

TC-ORD-001: Track Order Status in Real-Time

Field	Detail
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User Story	US-14: Track order status (processing, shipped, delivered)
Acceptance Criteria	Order status visible and updates in real-time
Priority	High
Test Type	Positive, Functional
Classification Tags	Regression

Preconditions:

User is logged in as frequent@test.com
User has placed order ORD-12345 within last hour
Order status in system: "Processing" initially

Test Data:

Order Field	Value
Order Number	ORD-12345
Initial Status	Processing
Later Status	Shipped, then Delivered

Test Steps:

Step	Action	Expected Result
1	Log into user account	Dashboard displayed
2	Navigate to "My Orders" page	Orders page loads
3	Locate order ORD-12345 in orders list	Order displayed with date, total, status
4	Verify initial status	Status shows "Processing"
5	Click on order to view details	Order detail page loads
6	Verify status indicator on detail page	Status shown prominently (green badge "Processing")

7	Simulate status change to "Shipped"	Status update triggered
8	Refresh My Orders page (within 30 seconds)	Status updates to "Shipped"
9	Verify shipping information appears	Tracking number appears (if available)
10	Click tracking number	Carrier tracking page opens
11	Verify estimated delivery date	Date displayed (e.g., "Expected delivery: April 15, 2025")
12	Simulate status change to "Delivered"	Status updates
13	Refresh order detail page	Status changes to "Delivered"
14	Verify delivery confirmation	"Delivered on [date]" message with checkmark
15	Check "Leave a Review" button	Button appears after delivery
16	Verify real-time update (no refresh)	Status updates automatically if WebSocket enabled

Postconditions:

Order status history logged with timestamps

User can view status anytime

Negative Variations:

TC-ORD-002: Order not found → "Order not found" message

TC-ORD-003: Tracking not available → "Tracking number will be available when shipped"

TC-ORD-004: Broken tracking link → "Tracking information unavailable" message

TC-ORD-005: View Order History for Reordering

Field	Detail
User Story	US-15: See order history to purchase same items again
Acceptance Criteria	All current and past orders visible with dates, items, totals
Priority	Medium
Test Type	Positive, Functional

Classification Tags	Regression
----------------------------	------------

Preconditions:

User is logged in as frequent@test.com
 User has 3 past orders over last 6 months
 Orders contain 1–3 items each

Test Data:

Order	Date	Items	Total	Status
ORD-12345	March 1, 2025	Design Patterns	\$54.99	Delivered
ORD-12340	February 15, 2025	Clean Code, Refactoring	\$89.98	Delivered
ORD-12335	January 10, 2025	Python Crash Course	\$29.99	Delivered

Test Steps:

Step	Action	Expected Result
1	Log into user account	Dashboard displayed
2	Click "My Orders"	Order history page loads
3	Verify page title	"Order History" displayed
4	Verify all 3 orders appear	3 orders displayed sorted by newest first
5	Check order ORD-12345	Correct details shown
6	Check order ORD-12340	Correct details shown
7	Check order ORD-12335	Correct details shown
8	Click order ORD-12340	Order detail page opens
9	Verify items in order	"Clean Code" and "Refactoring" displayed
10	Locate "Buy Again" button	Button visible for items or order
11	Click "Buy Again"	Item added to cart
12	Verify cart update	Cart contains selected item

13	Return to orders page	Navigation back works
14	Verify filter options	Filters visible (30 days, 6 months, etc.)
15	Apply "Last 30 days" filter	Only recent orders shown
16	Remove filter	All orders reappear
17	Search by order number "12340"	Only matching order displayed

Postconditions:

Order history persists for account lifetime
Reorder adds items to cart (no auto-purchase)

Negative Variations:

TC-ORD-006: No order history → "You haven't placed any orders yet" message
TC-ORD-007: Discontinued product → "Buy Again" disabled
TC-ORD-008: Price changed → Cart shows updated price with notification "Price changed since last purchase"

Test Suite 8: Device Support and Performance (DEV)

Classification: Responsiveness, Performance

TC-DEV-001: Responsive Design Across All Devices

Field	Detail
User Story	US-16: Use website on mobile, tablet, and laptop
Acceptance Criteria	Website responsive, buttons tappable, fonts readable
Priority	High
Test Type	Usability, Compatibility
Classification Tags	Responsiveness

Preconditions:

Website deployed to staging environment
Access to physical devices or BrowserStack
Test devices available: iPhone 14, Samsung S23, iPad Pro, Windows Laptop, MacBook

Test Data:

Device	Screen Size	Browser
iPhone 14	390x844	Safari
Samsung S23	360x780	Chrome
iPad Pro	1024x1366	Safari
Windows Laptop	1920x1080	Chrome, Firefox, Edge
MacBook	2560x1600	Safari, Chrome

Test Steps (All Devices):

Step	Action	Expected Result
1	Navigate to homepage	Homepage loads successfully
2	Verify horizontal scroll	No horizontal scroll appears
3	Check navigation menu	Mobile shows hamburger menu
4	Tap hamburger menu	Menu expands vertically
5	Tap navigation links	Navigation works and menu closes
6	Check search bar	Visible and usable at top
7	Perform search "fiction"	Search works correctly
8	Check book grid layout	Mobile shows 1–2 columns
9	Tap book card	Navigates to details page
10	Check button size	Buttons $\geq$ 44x44 pixels
11	Check font size	Minimum 16px readable text
12	Check input fields	Usable without zoom
13	Check tablet layout	2–3 column layout
14	Check laptop layout	4–5 column layout
15	Resize browser window	Layout adjusts responsively

16	Rotate device orientation	Layout adapts correctly
17	Test checkout flow	Forms usable on mobile
18	Verify images	No distortion on any device

Safari (iPhone/Mac) Checks:

Step	Action	Expected Result
19	Test Safari rendering	No layout issues
20	Test zoom behavior	Proper pinch/double tap zoom

Chrome Android Checks:

Step	Action	Expected Result
21	Test Chrome UI behavior	No UI breakage
22	Test back navigation	Works correctly

Postconditions:

Responsive behavior validated across devices

Device-specific issues logged

Performance Variations:

Test	Condition	Expected Result
TC-DEV-002	Slow 3G network	Basic content loads first, images lazy load
TC-DEV-003	Offline mode	Cached content or offline message shown

TC-DEV-004: Performance Load Testing - 1000 Concurrent Users

Field	Detail
User Story	US-20: Handle large traffic during peak hours
Acceptance Criteria	System supports 1000+ users, 99.9% availability

Priority	High
Test Type	Performance, Load
Classification Tags	Performance, Automation

Preconditions:

Performance environment is production clone
Load testing tool (JMeter) configured
1000 unique test users prepared
Monitoring tools active (CPU, memory, DB connections)

Test Data:

Metric	Target
Concurrent Users	1000
Test Duration	30 minutes
Ramp-up Time	5 minutes
User Flow	Login → Search → View Book → Add to Cart → View Cart → Logout

Test Steps:

Step	Action	Expected Result
1	Configure 1000 virtual users in JMeter	Test plan ready
2	Set ramp-up time to 5 minutes	Gradual user load
3	Define user journey	Login, search, browse, cart actions
4	Add think time (10–30 sec)	Realistic user behavior
5	Start load test	System begins handling traffic
6	Monitor CPU and memory	CPU ≤ 80%, Memory ≤ 85%
7	Monitor response times	Search ≤ 3 sec for 95% requests

8	Monitor book page load time	≤ 3 sec for 95% requests
9	Monitor error rate	$\leq 1\%$ errors
10	Run test for 30 minutes	System remains stable
11	Observe system recovery	Resources normalize within 5 min
12	Validate data integrity	No missing/duplicate orders
13	Check uptime	100% availability during test
14	Generate report	Load, throughput, response metrics shown
15	Run spike test	1000 users in 1 minute scenario

Postconditions:

Performance report generated

Bottlenecks identified for optimization

Scaling recommendations prepared

Acceptance Criteria Validation

Requirement	Target	Status
NFR-13: Concurrent users	≥ 1000	_____
NFR-7: Availability	99.9%	_____
Search response time	≤ 3 sec	_____
Book page load time	≤ 3 sec	_____
Checkout time	≤ 10 sec	_____

Negative Variations

Test Case	Scenario	Expected Result
TC-DEV-005	2000 concurrent users	System degrades gracefully, no crash
TC-DEV-006	Sudden spike (1000 users in 30 sec)	Auto-scaling triggers, temporary slowdown, recovery after stabilization

Test Suite Classification

The following table summarizes all 20+ test cases with their classifications for Testworthy organization:

Test Suite	Test Cases	Smoke	Regression	Negative	Security	Performance	Responsiveness	Automation Ready
Account Management	TC-ACC-001 to 014	✓	✓	✓	✓			✓
Search and Discovery	TC-SEARCH-001 to 009	✓	✓	✓		✓		✓
Book Information	TC-BOOK-001 to 008	✓	✓	✓		✓		✓
Wishlist	TC-WISH-001 to 008		✓	✓				✓
Shopping Cart	TC-CART-001 to 011	✓	✓	✓				✓

Checkout and Payment	TC-C HK-0 01 to 010	✓	✓	✓	✓	✓		
Order Managem ent	TC-O RD-0 01 to 008		✓	✓				✓
Device Support	TC-D EV-0 01 to 006		✓				✓	
Performa nce	TC-P ERF- 001 to 006					✓		✓

3. Creating Testworthy Project

Create New Project

Follow the wizard to set up your complete testing project

Step 2 of 9

Project Details

✓ Setup Method

2 Project Details

3 Milestones

4 Test Plans

5 Test Suites

6 Sections

7 Test Cases

Project Details

Basic information about your project

Project Name *

Online Book Store

Description

This project encompasses the test planning and execution for the Online Book Store application. It validates all functional and non-functional requirements, including user stories for account management, search and discovery, book information, wishlist, shopping cart, checkout and payment, order management, and device support. The goal is to ensure a high-quality, secure, and performant e-commerce experience.

Project Participants

Enter email addresses of team members (comma-separated)

Cancel

← Back

Next →

4. Defining Milestones (Sprints 1-7)

Create New Project

Follow the wizard to set up your complete testing project

Name	Description	Due Date	Actions
Sprint 1 – Account Management	Validate user registration (US-1), login (US-2), session timeout after 1 hour (US-18), and password encryption (NFR-1). Includes smoke, functional, negative, security, and regression testing.	2026-05-08	
Sprint 2 – Search and Discovery	Validate search by title/author/category (FR-1, US-3), filters by price/language/year/rating (FR-3, US-4), sort by price (FR-4, US-5), and search result loading within 3 seconds (NFR-4, US-19).	2026-05-15	
Sprint 3 – Book Information	Validate complete book details display (FR-2, US-6), user reviews and ratings (FR-6, US-7), and book detail page loading within 3 seconds (NFR-5).	2026-05-22	
Sprint 4 – Wishlist and Shopping Cart	Validate add/remove books from wishlist (FR-5, US-8), add/remove items from cart (FR-8, US-9), and grand total calculation (FR-8, US-10).	2026-05-29	
Sprint 5 – Checkout and Payment	Validate secure PCI DSS compliant payment (FR-7, US-11), save address/payment info (FR-10, US-12), order confirmation email (FR-12, US-13), checkout within 10 seconds (NFR-6), and email delivery within 5 minutes (NFR-11).	2026-06-05	
Sprint 6 – Order Management	Validate order tracking with status updates (FR-9, US-14), order history retention (FR-11, US-15), and real-time status display (NFR-12).	2026-06-12	
Sprint 7 – Device Support and	Validate responsive design across devices (FR-13, NFR-8, US-16), legible fonts (NFR-9), tappable buttons (NFR-10), support for 1000+ concurrent	2026-	

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5. Adding Test Plans

Create New Project



Follow the wizard to set up your complete testing project

Test Plan – Sprint 1: Account Management	Validates user registration (US-1), login (US-2), session timeout (US-18), and password encryption (NFR-1). Includes smoke, functional, negative, security, and regression test cases.	Sprint 1 – Account Management	
Test Plan – Sprint 2: Search and Discovery	Validates search by title/author/category (FR-1, US-3), filters (FR-3, US-4), sort by price (FR-4, US-5), and search loading within 3 sec (NFR-4, US-19).	Sprint 2 – Search and Discovery	
Test Plan – Sprint 3: Book Information	Validates complete book details (FR-2, US-6), user reviews (FR-6, US-7), and book page load within 3 sec (NFR-5).	Sprint 3 – Book Information	
Test Plan – Sprint 4: Wishlist and Shopping Cart	Validates wishlist add/remove (FR-5, US-8), cart add/remove (FR-8, US-9), and grand total calculation (FR-8, US-10).	Sprint 4 – Wishlist and Shopping Cart	
Test Plan – Sprint 5: Checkout and Payment	Validates secure PCI DSS payment (FR-7, US-11), save address/payment (FR-10, US-12), order email (FR-12, US-13), checkout within 10 sec (NFR-6), email within 5 min (NFR-11).	Sprint 5 – Checkout and Payment	
Test Plan – Sprint 6: Order Management	Validates order tracking (FR-9, US-14), order history (FR-11, US-15), and real-time status (NFR-12).	Sprint 6 – Order Management	
Test Plan – Sprint 7: Device Support and Performance	Validates responsive design (FR-13, NFR-8, US-16), legible fonts (NFR-9), tappable buttons (NFR-10), 1000+ concurrent users (NFR-13, US-20), and 99.9% availability (NFR-7).	Sprint 7 – Device Support and Performance	

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6. Creating Test Suites (8 suites)

Create New Project

Follow the wizard to set up your complete testing project

Step 5 of 9

Test Suites

✓ Setup Method ✓ Project Details ✓ Milestones ✓ Test Plans **5 Test Suites** 6 Sections 7 Test Cases

Test Suites

Organize tests into logical suites

Add Test Suite

Name	Description	Test Plan	Actions
Suite 1: Account Management (ACC)	Validates user registration (US-1), login (US-2), session timeout (US-18), and password encryption (NFR-1). Contains test cases for smoke, functional, negative, security, and regression testing.	Test Plan – Sprint 1: Account Management	
Suite 2: Search and Discovery (SRCH)	Validates search by title/author/category (FR-1, US-3), filters by price/language/year/rating (FR-3, US-4), sort by price (FR-4, US-5), and search loading within 3 sec (NFR-4, US-19).	Test Plan – Sprint 2: Search and Discovery	
Suite 3: Book Information (BOOK)	Validates complete book details display (FR-2, US-6), user reviews/ratings (FR-6, US-7), and book page loading within 3 sec (NFR-5).	Test Plan – Sprint 3: Book Information	
	Validates saving books to wishlist for future consideration	Test Plan – Sprint 4:	

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7. Creating Test Cases

Create New Project

Follow the wizard to set up your complete testing project

Project Summary

Project Information

Name: Online Book Store

Description: This project encompasses the test planning and execution for the Online Book Store application. It validates all functional and non-functional requirements, including user stories for account management, search and discovery, book information, wishlist, shopping cart, checkout and payment, order management, and device support. The goal is to ensure a high-quality, secure, and performant e-commerce experience.

Setup Type: Manual Setup

7

Milestones

7

Test Plans

8

Test Suites

23

Sections

77

Test Cases

7

Test Runs

Relationship Summary

Hierarchy: 7 Milestones → 7 Test Plans → 8 Test Suites → 23 Sections → 77 Test Cases → 7 Test Runs

Linkages: 7/7 Test Plans to Milestones • 77/77 Test Cases in Suites • 7/7 Test Runs to Plans • 7/7 Test Runs to Milestones

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**Test Case Management With
Testworthy**

1 May 2026



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